Topic: Variation of parameters for nonhomogeneous equations

Question: Find the general solution to the differential equation.

$$y'' - 2y' + y = \frac{3e^x}{2x}$$

Answer choices:

A $y = c_1 e^x + c_2 x e^x + \frac{3}{2} e^x \ln|x|$

B $y(x) = c_1 e^x + c_2 x e^x + \frac{3}{2} x \ln|x|$

C $y(x) = c_1 e^x + c_2 x e^x + \frac{3}{2} x e^x - \frac{3}{2} x e^x \ln|x|$

D $y(x) = c_1 e^x + c_2 x e^x - \frac{3}{2} x e^x + \frac{3}{2} x e^x \ln|x|$

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Solution: D

The coefficient on y'' is already 1. We'll substitute $g(x) = 0$ to rewrite the nonhomogeneous equation as a homogeneous equation.

$$y'' - 2y' + y = 0$$

Now we'll use the associated characteristic equation to find roots.

$$r^2 - 2r + 1 = 0$$

$$(r-1)(r-1) = 0$$

$$r = 1$$

We get equal real roots, so the complementary solution is

$$y_c(x) = c_1 e^x + c_2 x e^x$$

and the fundamental set of solutions and their derivatives are therefore

$$\{y_1, y_2\} = \{e^x, xe^x\}$$

$$\{y_1', y_2'\} = \{e^x, e^x + xe^x\}$$

Create a system of linear equations using the fundamental set and its derivatives.

$$u_1' y_1 + u_2' y_2 = 0 \qquad u_1' y_1' + u_2' y_2' = g(x)$$

$$u_1' e^x + u_2' x e^x = 0 \qquad u_1' e^x + u_2'(e^x + xe^x) = \frac{3e^x}{2x}$$

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$$u_1'e^x + u_2'e^x + u_2'xe^x = \frac{3e^x}{2x}$$

Subtract one equation from the other.

$$(u_1'e^x + u_2'xe^x) - (u_1'e^x + u_2'e^x + u_2'xe^x) = 0 - \frac{3e^x}{2x}$$

$$-u_2'e^x = -\frac{3e^x}{2x}$$

$$u_2' = \frac{3}{2x}$$

Use this value for u_2' to find u_1'

$$u_1'e^x + \left(\frac{3}{2x}\right)xe^x = 0$$

$$u_1'e^x = -\frac{3}{2}e^x$$

$$u_1' = -\frac{3}{2}$$

Integrate u_1' and u_2' to find u_1 and u_2.

$$u_1 = \int -\frac{3}{2}\,dx \qquad u_2 = \int \frac{3}{2x}\,dx$$

$$u_1 = -\frac{3}{2}x \qquad u_2 = \frac{3}{2}\ln|x|$$

So the particular solution is

$$y_p(x) = u_1y_1 + u_2y_2$$

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$$y_p(x) = -\frac{3}{2}x(e^x) + \frac{3}{2}\ln|x|(xe^x)$$

$$y_p(x) = -\frac{3}{2}xe^x + \frac{3}{2}xe^x \ln|x|$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1e^x + c_2xe^x - \frac{3}{2}xe^x + \frac{3}{2}xe^x \ln|x|$$

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Topic: Variation of parameters for nonhomogeneous equations

Question: Find the general solution to the differential equation.

$$y'' - 3y' - 4y = e^{3x}$$

Answer choices:

A $y(x) = c_1 e^{-x} + c_2 e^{4x} - \dfrac{e^{3x}}{4}$

B $y(x) = c_1 e^{x} + c_2 e^{-4x} + \dfrac{e^{3x}}{4}$

C $y(x) = c_1 e^{-x} + c_2 x e^{4x} - \dfrac{e^{3x}}{4}$

D $y(x) = c_1 e^{x} + c_2 x e^{-4x} + \dfrac{e^{3x}}{4}$

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Solution: A

The coefficient on y'' is already 1. We'll substitute $g(x) = 0$ to rewrite the nonhomogeneous equation as a homogeneous equation.

$$y'' - 3y' - 4y = 0$$

Now we'll use the associated characteristic equation to find roots.

$$r^2 - 3r - 4 = 0$$

$$(r - 4)(r + 1) = 0$$

$$r = 4, \ -1$$

We get distinct real roots, so the complementary solution is

$$y_c(x) = c_1 e^{-x} + c_2 e^{4x}$$

and the fundamental set of solutions and their derivatives are therefore

$$\{y_1, y_2\} = \{e^{-x}, e^{4x}\}$$

$$\{y_1', y_2'\} = \{-e^{-x}, 4e^{4x}\}$$

Create a system of linear equations using the fundamental set and its derivatives.

$$u_1' y_1 + u_2' y_2 = 0$$

$$u_1' e^{-x} + u_2' e^{4x} = 0$$

$$u_1' y_1' + u_2' y_2' = g(x)$$

$$u_1'(-e^{-x}) + u_2'(4e^{4x}) = e^{3x}$$

$$-u_1' e^{-x} + 4u_2' e^{4x} = e^{3x}$$

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Add the equations.

$$(u_1'e^{-x} + u_2'e^{4x}) + (-u_1'e^{-x} + 4u_2'e^{4x}) = 0 + e^{3x}$$

$$5u_2'e^{4x} = e^{3x}$$

$$u_2' = \frac{e^{-x}}{5}$$

Use this value for u_2' to find u_1'

$$u_1'e^{-x} + \left(\frac{e^{-x}}{5}\right)e^{4x} = 0$$

$$u_1'e^{-x} = -\frac{e^{3x}}{5}$$

$$u_1' = -\frac{e^{4x}}{5}$$

Integrate u_1' and u_2' to find u_1 and u_2.

$$u_1 = \int -\frac{e^{4x}}{5}\,dx \qquad u_2 = \int \frac{e^{-x}}{5}\,dx$$

$$u_1 = -\frac{e^{4x}}{20} \qquad u_2 = -\frac{e^{-x}}{5}$$

So the particular solution is

$$y_p(x) = u_1y_1 + u_2y_2$$

$$y_p(x) = -\frac{e^{4x}}{20}(e^{-x}) - \frac{e^{-x}}{5}(e^{4x})$$

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$$y_p(x) = -\frac{e^{3x}}{20} - \frac{e^{3x}}{5}$$

$$y_p(x) = -\frac{e^{3x}}{4}$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = c_1 e^{-x} + c_2 e^{4x} - \frac{e^{3x}}{4}$$

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Topic: Variation of parameters for nonhomogeneous equations

Question: Find the general solution to the differential equation.

$$y'' + 2y' + 2y = e^{-x}\cos^2 x$$

Answer choices:

A $$y(x) = e^{-x}\left(c_1 \cos x + c_2 \sin x + \frac{1}{3}\cos^2 x\right)$$

B $$y(x) = e^{-x}\left(c_1 \cos x + c_2 \sin x + \frac{1}{3}\sin^2 x\right)$$

C $$y(x) = e^{-x}\left(c_1 \cos x + c_2 \sin x + \frac{1}{3} + \frac{1}{3}\cos^2 x\right)$$

D $$y(x) = e^{-x}\left(c_1 \cos x + c_2 \sin x + \frac{1}{3} + \frac{1}{3}\sin^2 x\right)$$

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Solution: D

The coefficient on y'' is already 1. We'll substitute $g(x) = 0$ to rewrite the nonhomogeneous equation as a homogeneous equation.

$$y'' + 2y' + 2y = 0$$

Now we'll use the associated characteristic equation to find roots.

$$r^2 + 2r + 2 = 0$$

$$(r + 1)^2 + 1 = 0$$

$$r = -1 \pm i$$

We get two complex roots, so the complementary solution is

$$y_c(x) = e^{-x}(c_1 \cos x + c_2 \sin x)$$

and the fundamental set of solutions and their derivatives are therefore

$$\{y_1, y_2\} = \{e^{-x} \cos x, e^{-x} \sin x\}$$

$$\{y_1', y_2'\} = \{-e^{-x} \cos x - e^{-x} \sin x, -e^{-x} \sin x + e^{-x} \cos x\}$$

Create a system of linear equations using the fundamental set and its derivatives. We find

$$u_1' y_1 + u_2' y_2 = 0$$

$$u_1' e^{-x} \cos x + u_2' e^{-x} \sin x = 0$$

$$u_1' \cos x + u_2' \sin x = 0$$

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and

$$u_1'y_1' + u_2'y_2' = g(x)$$

$$u_1'e^{-x}(-\cos x - \sin x) + u_2'e^{-x}(-\sin x + \cos x) = e^{-x}\cos^2 x$$

$$u_1'(-\cos x - \sin x) + u_2'(-\sin x + \cos x) = \cos^2 x$$

Add the equations.

$$-u_1' \sin x + u_2' \cos x = \cos^2 x$$

From first equation, we know

$$u_2' = -u_1' \frac{\cos x}{\sin x}$$

and we can substitute this into the sum of the equations to get

$$-u_1' \sin x + \left(-u_1' \frac{\cos x}{\sin x}\right)' \cos x = \cos^2 x$$

$$-u_1' \frac{\sin^2 x + \cos^2 x}{\sin x} = \cos^2 x$$

$$-u_1' \frac{1}{\sin x} = \cos^2 x$$

$$u_1' = -\sin x \cos^2 x$$

Use this value for u_1' to find u_2'.

$$u_2' = -(-\sin x \cos^2 x)\frac{\cos x}{\sin x}$$

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$$u_2' = \cos^3 x$$

Integrate u_1' and u_2' to find u_1 and u_2. We get

$$u_1 = \int -\sin x \cos^2 x \ dx$$

$u = \cos x$ and $du = -\sin x \ dx$

$$u_1 = \int u^2 \ du$$

$$u_1 = \frac{1}{3}u^3 = \frac{1}{3}\cos^3 x$$

and

$$u_2 = \int \cos^3 x \ dx$$

$$u_2 = \int (1 - \sin^2 x)\cos x \ dx$$

$u = \sin x$ and $du = \cos x \ dx$

$$u_2 = \int 1 - u^2 \ du$$

$$u_2 = u - \frac{1}{3}u^3 = \sin x - \frac{1}{3}\sin^3 x$$

So the particular solution is

$$y_p(x) = u_1 y_1 + u_2 y_2$$

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$$y_p(x) = \frac{\cos^3 x}{3}(e^{-x}\cos x) + \left(\sin x - \frac{\sin^3 x}{3}\right)(e^{-x}\sin x)$$

$$y_p(x) = \frac{\cos^4 x}{3}e^{-x} + \left(\sin^2 x - \frac{\sin^4 x}{3}\right)e^{-x}$$

$$y_p(x) = \left(\frac{\cos^4 x - \sin^4 x + 3\sin^2 x}{3}\right)e^{-x}$$

$$y_p(x) = \left(\frac{(1-\sin^2 x)^2 - \sin^4 x + 3\sin^2 x}{3}\right)e^{-x}$$

$$y_p(x) = \left(\frac{1 - 2\sin^2 x + \sin^4 x - \sin^4 x + 3\sin^2 x}{3}\right)e^{-x}$$

$$y_p(x) = \frac{1+\sin^2 x}{3}e^{-x}$$

Adding this particular solution to the complementary solution gives us the general solution $y(x)$.

$$y(x) = e^{-x}\left(c_1\cos x + c_2\sin x + \frac{1}{3} + \frac{1}{3}\sin^2 x\right)$$

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